Seoyul Jeon

jsy628496@kaist.ac.kr github.com/0x4A656F6E2053656F79756C
+82-10-7570-2351 0x4A656F6E2053656F79756C.github.io

Education

Mar 2021 – Present

B.S. in Computer Science (Minor in Electrical Engineering)

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, S.Korea

- Total GPA: 3.66 / 4.3
- Credits taken: 151

Jan 2024 – Jul 2024

Exchange Student

Lund University, Lund, Sweden

- Faculty of Engineering, Department of Computer Science
- Courses taken: Artificial Intelligence, Numerical Analysis, Advanced Algorithms, Functional Programming

Research Experience

Jan 2026 – Present

Research Intern

Sensing Intelligence and Cyber-Physical Security (CyPhy) Lab

Wi-Fi CSI-based Presence Detection System

- Researched algorithms for detecting human presence using Channel State Information (CSI) based on the IEEE 802.11bf standard.
- Built a data collection environment using Raspberry Pi 4B and AX210 wireless network cards, and developed the signal processing pipeline in Python.

Fall 2025

CS Project

Graduation Research

Deep Autoencoder-based Network Anomaly Detection System

- Implemented a deep autoencoder model for network anomaly detection from scratch in pure C++ without using external ML libraries.
- Processed the UNSW-NB15 dataset and achieved 99.08% recall and 97.24% accuracy on 1.84 million unseen network traffic samples.

Winter 2024

Individual Research

Flexible Machine Learning (FlexML) Lab

AlphaZero-based AI Engine Implementation & LLM Alignment

- Analyzed and implemented Monte Carlo Tree Search (MCTS) algorithm and Two-headed Neural Network architectures to understand strategic planning in game-playing AI.
- Studied preference-based alignment techniques, including Direct Preference Optimization (DPO), for aligning Large Language Models without explicit reward models.

Academic Projects

Feb 2026 – Present

User Identification via Remote Control IMU

MobiSys Poster Presentation Project (Co-first author)

- Developed a user identification system utilizing the LSM6DSO IMU sensor embedded in remote controls.
- Processed motion data and applied machine learning techniques to classify unique user interaction patterns.

Spring 2022

Computer Vision Barrier-free Kiosk

Social Innovation Lab with Technology Course (Team Leader)

- Developed an image processing pipeline using AI to determine users' disabilities via computer vision.
- Implemented hardware control logic for the kiosk within an embedded systems environment.

Awards & Honors

Dec 2022 — **Encouragement Award**, IBS Basic Science Promotional Content Contest

2021 - 2023 — **Presidential Science Scholarship**, Awarded to promising future science and engineering talents in Korea

Skills

Programming	Python, C/C++
Hardware/Systems	Raspberry Pi, Arduino, Linux, Wireless Network cards, IMU
Languages	English (TOEFL iBT: 83, Mar 2023), Korean (Native)